DigitalOcean hosted inference

Technical benchmark and engineering guide for the 11 DigitalOcean-hosted open-model routes frozen in the 27-29 August 2026 study

11	1,232	7	92.1%
hosted open-model endpoints	matched low-load requests	hourly panels over six hours	overall request success

Bottom line: eight endpoints completed every request in the registered six-hour low-load panel. Qwen3.5 397B A17B and Nemotron 3 Super 120B became materially rate-limited late in the run; Gemma 4 31B had smaller reliability losses. Capacity and fixed-rate results remain workload-specific.

This report separates observed behavior from product documentation, reports uncertainty and denominators, and never converts an unmeasured or failed condition into a blank success. Commercial pass-through routes are outside this DigitalOcean-hosted study.

Measurement window: 28-29 August 2026 (UTC) | Publication build: 29 August 2026

Claim boundary: results characterize this measured configuration and six-hour within-run window. The catalog can change after this dated snapshot. These results are not a 24-hour, daily, diurnal, or indefinite production guarantee.

Engineering decision map

Start with reliability at low offered load, then consult the exact workload page for capacity. A green reliability result does not imply that every feature or high-load workload is suitable.

Endpoint	6 h low-load reliability	Adaptive-load cells with repeat-confirmed result	2 min fixed-rate passes	Engineering reading
DeepSeek V4 Flash	112/112 = 100.0% 95% CI 96.7-100.0%	4/4	1/4	Reliable at the registered low-load panel; size with the exact workload evidence.
Gemma 4 31B	105/112 = 93.8% 95% CI 87.7-96.9%	3/4	1/4	Mostly reliable here; retain retry/backoff and watch late-run errors.
GLM 5.2	112/112 = 100.0% 95% CI 96.7-100.0%	1/4	0/4	Reliable at the registered low-load panel; size with the exact workload evidence.
Kimi K3	112/112 = 100.0% 95% CI 96.7-100.0%	1/4	0/4	Reliable at the registered low-load panel; size with the exact workload evidence.
MiniMax M2.5	112/112 = 100.0% 95% CI 96.7-100.0%	3/4	0/4	Reliable at the registered low-load panel; size with the exact workload evidence.
MiMo V2.5 Pro	112/112 = 100.0% 95% CI 96.7-100.0%	2/4	0/4	Reliable at the registered low-load panel; size with the exact workload evidence.
Nemotron 3 Ultra 550B	112/112 = 100.0% 95% CI 96.7-100.0%	3/4	0/4	Reliable at the registered low-load panel; size with the exact workload evidence.
Nemotron 3 Super 120B	62/112 = 55.4% 95% CI 46.1-64.2%	0/4	0/4	Material rate limiting in this run; use conservative concurrency and live telemetry.
GPT-OSS 120B	112/112 = 100.0% 95% CI 96.7-100.0%	3/4	0/4	Reliable at the registered low-load panel; size with the exact workload evidence.
Qwen3.5 397B A17B	72/112 = 64.3% 95% CI 55.1-72.6%	0/4	0/4	Material rate limiting in this run; use conservative concurrency and live telemetry.
Qwen3.8 Max	112/112 = 100.0% 95% CI 96.7-100.0%	4/4	1/4	Reliable at the registered low-load panel; size with the exact workload evidence.

Across all endpoints: 1135/1232 successes (Wilson 95% CI 90.49-93.50%). Do not average endpoint latency or throughput into a provider ranking; models and recipes differ.

Exactly what was tested

Four workload recipes exercise different bottlenecks. Every result in this report stays attached to its endpoint, recipe, source campaign, and sampling unit.

Recipe	Registered request	What it stresses	How to use the result
Short / short	256-token prompt -> 128-token answer target	Request scheduling, queueing, and first-token response	Interactive assistants and small structured calls
Long input / short answer	100,000-token prompt -> 128 tokens; MiniMax uses 50,000 tokens	Prompt ingestion, context handling, and cache reuse	Document analysis and retrieval-heavy requests
Short / long	256-token prompt -> 4,096-token answer target	Long generation, request occupancy, timeouts	Drafting, synthesis, and extended reasoning outputs
Mixed traffic	Fixed seeded blend of short, long-input, long-output, JSON, and tool-like requests	Scheduler fairness across a realistic heterogeneous queue	Shared application traffic; do not compare its latency to one homogeneous recipe

Experiment sequence

1. Functional and boundary probes establish what the API accepts. 2. Adaptive load search raises offered traffic, backs off on degradation, and requires three separated healthy confirmations. 3. A two-minute fixed-rate test holds one selected rate for four adjacent 30-second blocks and checks recovery. 4. Seven matched hourly panels measure low-load variation over exactly six hours.

Internal code name note: the adaptive load search is AIMD (additive increase, multiplicative decrease). The two-minute fixed-rate stability test was historically called a soak; this report uses the plain name.

Evidence and uncertainty

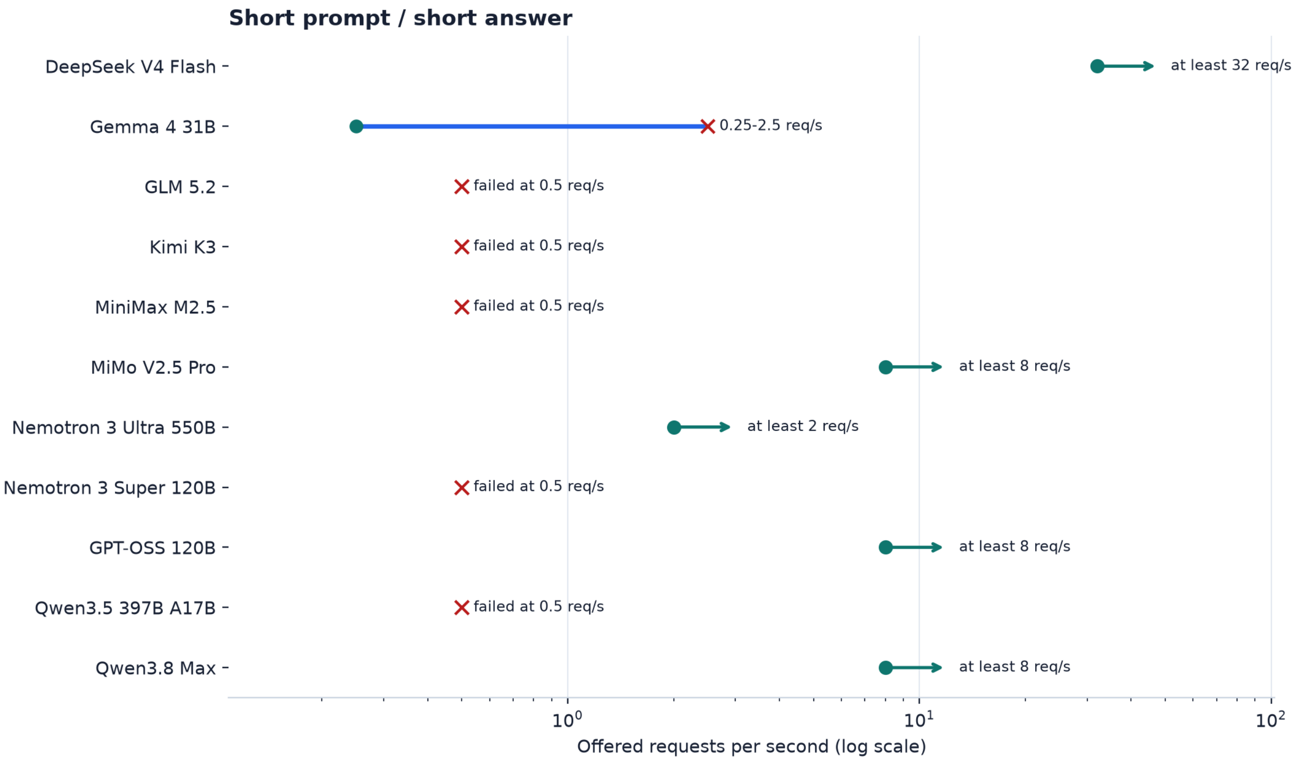
Question	Evidence	Uncertainty unit	Valid claim
Does a request work?	Functional/capability probes	Request; Wilson interval for binary outcomes	Observed support or measured failure for the exact request
What rate repeatedly passes?	Adaptive load search + three separated confirmations	Load epoch/block	Measured lower bound or bracket, never a theoretical maximum
Does one rate hold briefly?	Four contiguous 30-second fixed-rate blocks	Block; exploratory Student-t interval	Two-minute measured pass/failure at that rate
Did behavior change over six hours?	Seven matched hourly panels	Request-level Wilson/bootstrap; panel-cluster intervals for repeated-panel effects	Within-run six-hour variation only

Metric eligibility

End-to-end latency is reported for successful requests. Time to first token is reported only when the stream exposed a valid first-output timestamp. Decode throughput additionally requires complete usage, a decode window of at least one second, and at least 16 output tokens. In the six-hour panel only 198 requests met that stricter throughput rule, across three endpoints; the other eight endpoints are labelled not estimable rather than assigned a number.

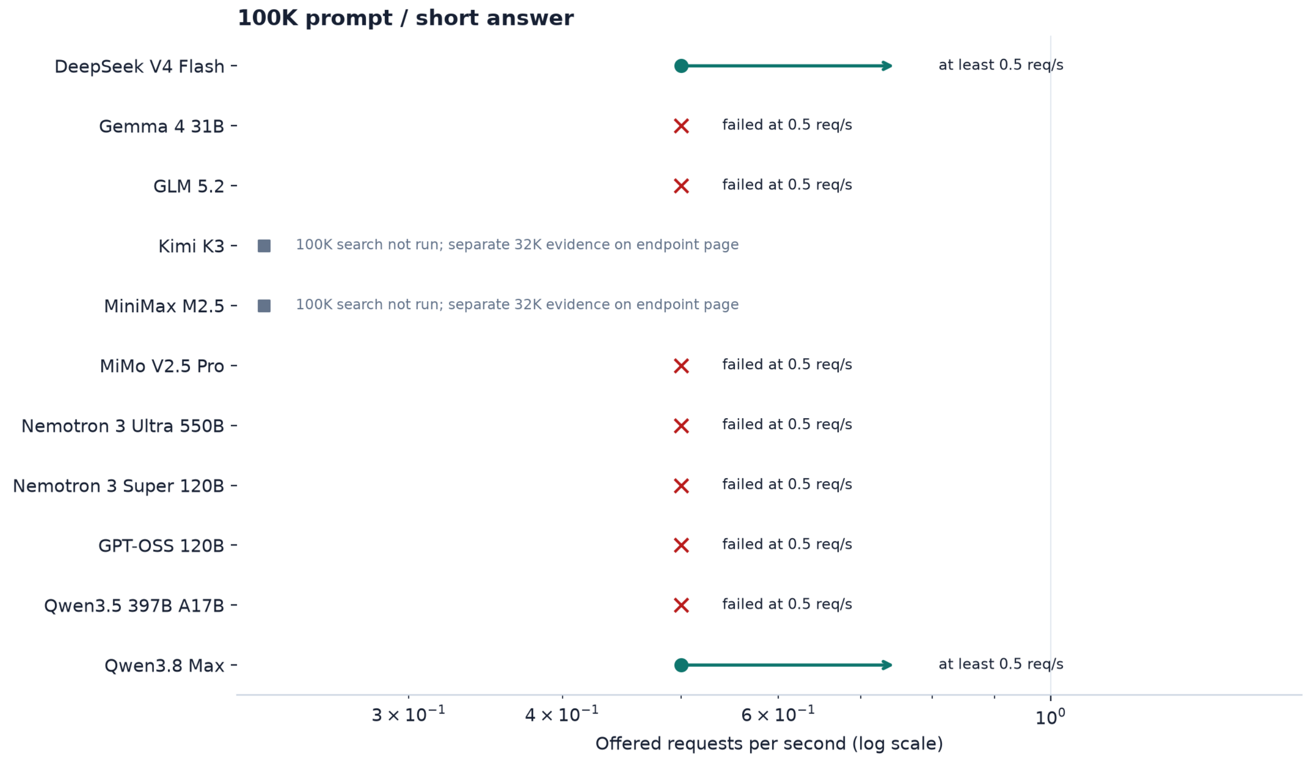
The eligible six-hour decode observations ranged up to 87.3 tokens/s. Impossible historical outliers are excluded by the explicit timestamp, duration, output-length, and usage checks and retained in the audit.

Adaptive load search: Short prompt / short answer



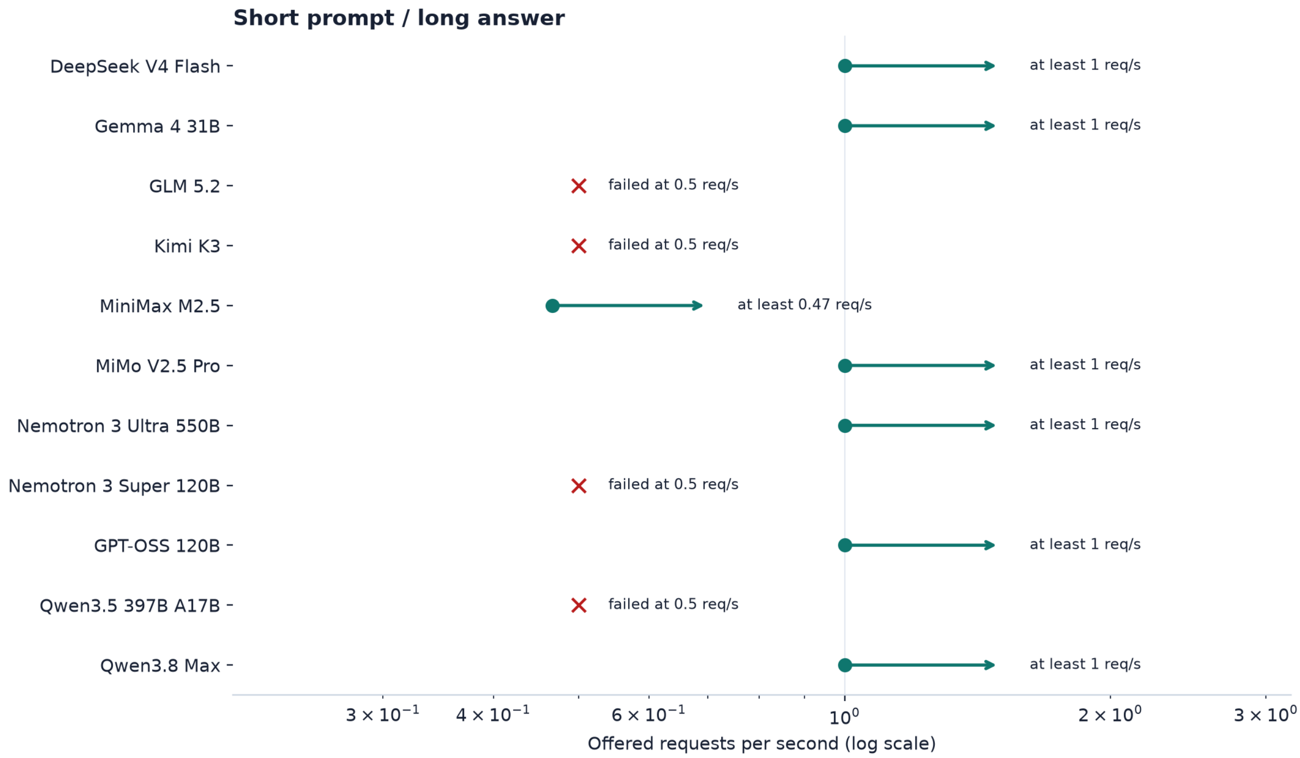
Tested	Open-loop offered traffic for this exact endpoint and recipe; a numeric result requires three separated healthy confirmations.
Shows	A teal point is a repeatedly passing tested rate; an orange/red marker is a measured limit, failure, or non-confirmed state.
Does not show	Theoretical maximum, recommended production rate, or performance for a different prompt/output recipe.

Adaptive load search: 100K prompt / short answer



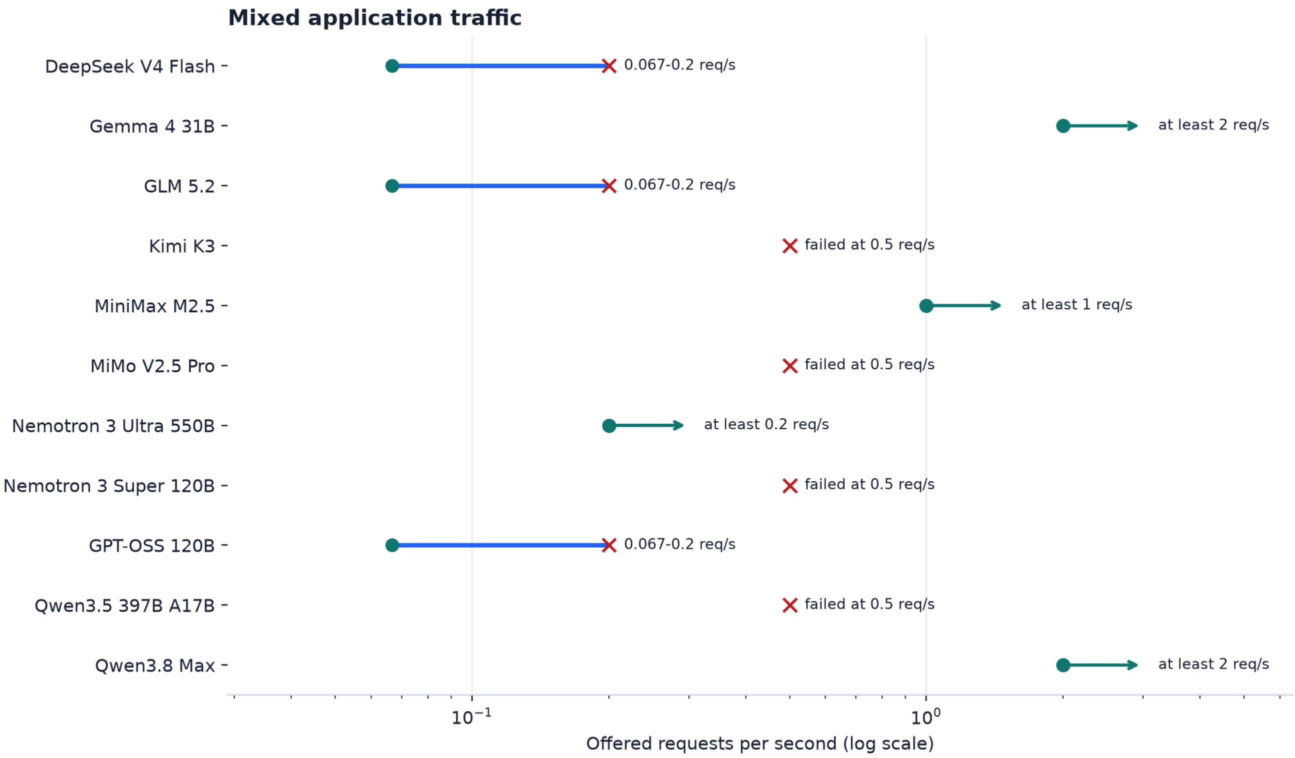
Tested	Open-loop offered traffic for this exact endpoint and recipe; a numeric result requires three separated healthy confirmations.
Shows	A teal point is a repeatedly passing tested rate; an orange/red marker is a measured limit, failure, or non-confirmed state.
Does not show	Theoretical maximum, recommended production rate, or performance for a different prompt/output recipe.

Adaptive load search: Short prompt / long answer



Tested	Open-loop offered traffic for this exact endpoint and recipe; a numeric result requires three separated healthy confirmations.
Shows	A teal point is a repeatedly passing tested rate; an orange/red marker is a measured limit, failure, or non-confirmed state.
Does not show	Theoretical maximum, recommended production rate, or performance for a different prompt/output recipe.

Adaptive load search: Mixed application traffic



Tested	Open-loop offered traffic for this exact endpoint and recipe; a numeric result requires three separated healthy confirmations.
Shows	A teal point is a repeatedly passing tested rate; an orange/red marker is a measured limit, failure, or non-confirmed state.
Does not show	Theoretical maximum, recommended production rate, or performance for a different prompt/output recipe.

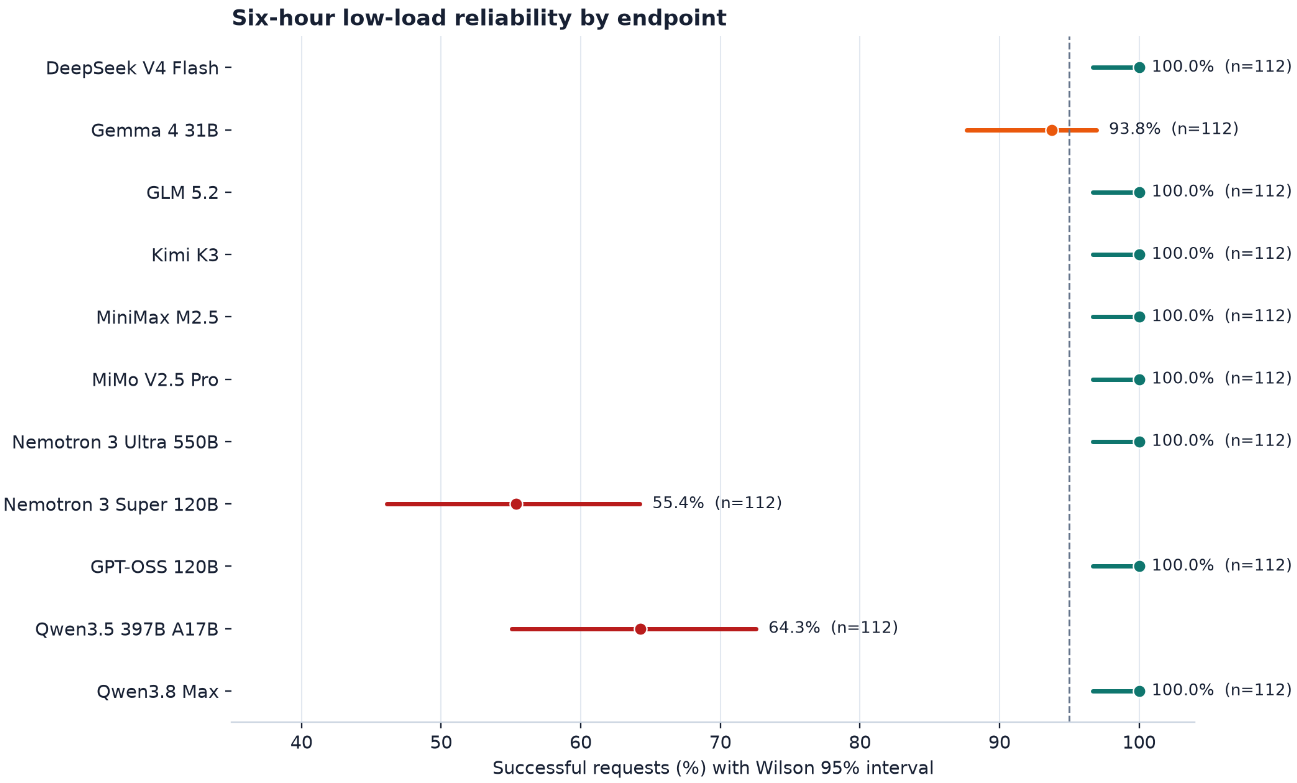
Two-minute fixed-rate stability

Two-minute fixed-rate stability: registered acceptance result

	Short prompt / short answer	32K prompt / short answer	Short prompt / long answer	Mixed application traffic
DeepSeek V4 Flash	PASS	FAIL	FAIL	FAIL
Gemma 4 31B	PASS	FAIL	FAIL	FAIL
GLM 5.2	FAIL	FAIL	FAIL	FAIL
Kimi K3	FAIL	FAIL	FAIL	FAIL
MiniMax M2.5	FAIL	FAIL	FAIL	FAIL
MiMo V2.5 Pro	FAIL	FAIL	FAIL	FAIL
Nemotron 3 Ultra 550B	FAIL	FAIL	FAIL	FAIL
Nemotron 3 Super 120B	FAIL	FAIL	FAIL	FAIL
GPT-OSS 120B	FAIL	FAIL	FAIL	FAIL
Qwen3.5 397B A17B	FAIL	FAIL	FAIL	FAIL
Qwen3.8 Max	PASS	FAIL	FAIL	FAIL

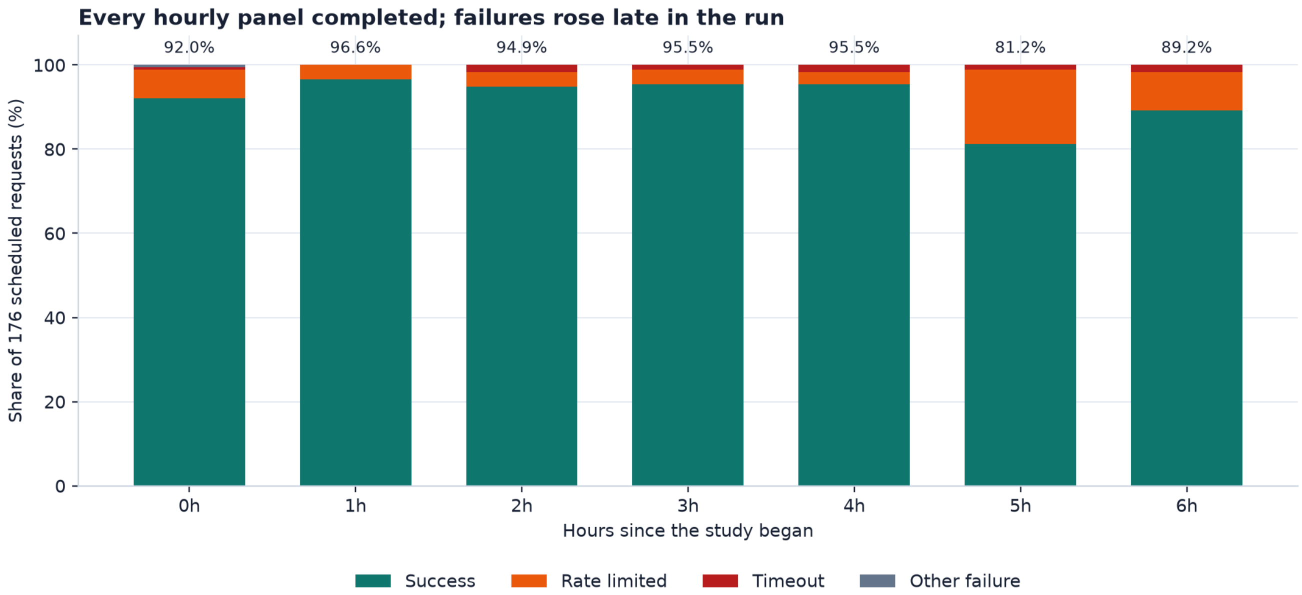
Tested	One candidate rate for four adjacent 30-second analysis blocks, with registered reliability, latency, queueing, usage, quality, and recovery checks.
Shows	3 of 44 endpoint-by-workload cells passed every acceptance condition; measured failures are evidence, not missing data.
Does not show	Six-hour stability or safe operation above the tested rate. Adjacent-block intervals do not model serial correlation.

Six-hour matched panel: reliability



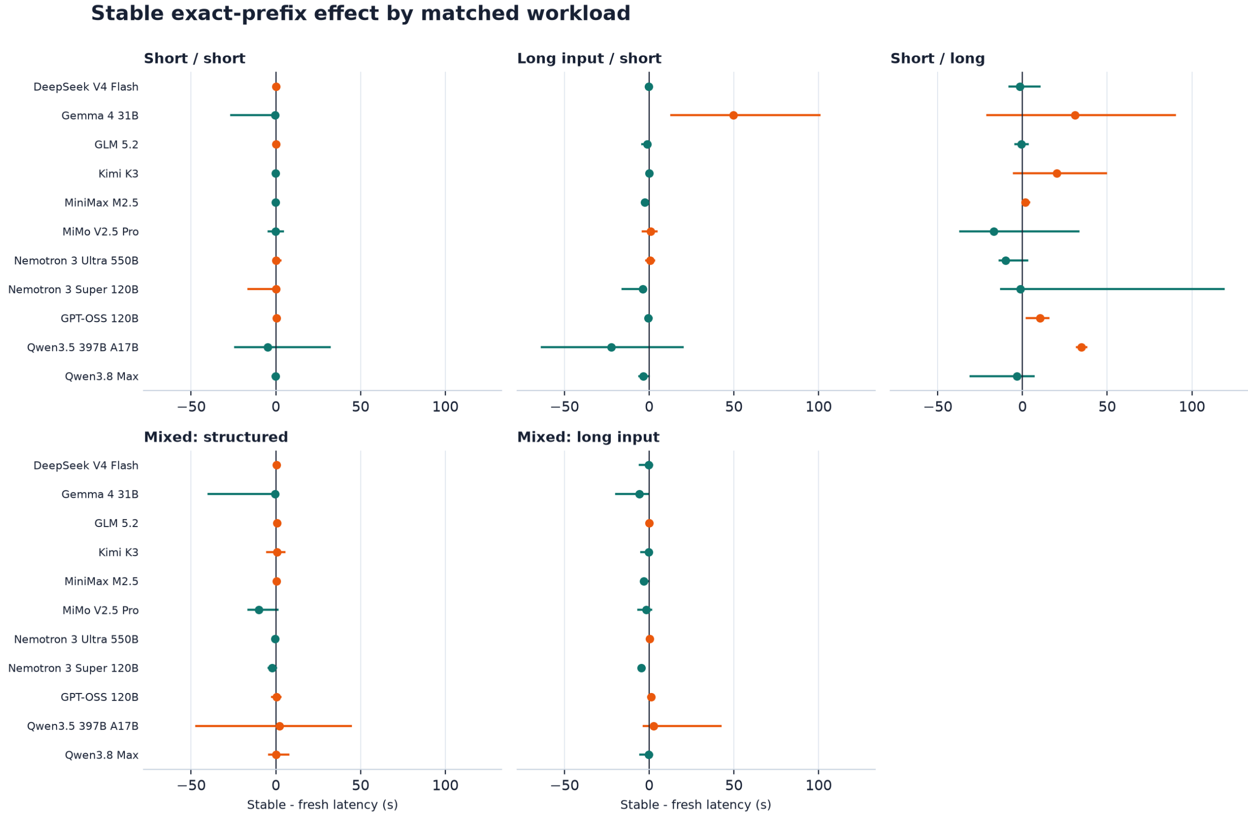
Tested	112 scheduled low-load requests per endpoint: four recipes x four repeats x seven hourly panels; no retries.
Shows	Endpoint-specific success rate with Wilson 95% interval. Eight endpoints completed all 112 requests.
Does not show	Behavior at high concurrency, outside this six-hour window, or under a different region/account/quota state.

Six-hour matched panel: when failures appeared



Tested	Seven globally scheduled panels at hours 0-6; each panel contains 176 matched requests at one request per second.
Shows	All panels completed. Success fell in hours 5 and 6, driven mainly by HTTP 429 responses on three endpoints.
Does not show	A daily or diurnal pattern. Seven points from one contiguous run cannot separate time-of-day from transient service or quota conditions.

Prompt reuse and automatic exact-prefix caching



Tested	Within each endpoint, recipe, and panel, two stable exact-prefix requests were paired with two panel-unique fresh-prefix requests.
Shows	Effects are reported separately for each endpoint and recipe. Some cells were faster with a stable prefix; many uncertainty intervals crossed zero.
Does not show	A cache hit guarantee or universal latency reduction. DigitalOcean caching is automatic and best effort.

Stable and fresh-prefix reliability was similar overall, but combining unlike models and recipes into one effect would hide important differences. The chart and paired-cache table therefore keep every endpoint and recipe separate. Negative values mean the stable prefix was faster; an interval crossing zero means this six-hour run could not establish the direction. Cache-read counters confirm reuse only where the provider exposed it; a cache read does not guarantee lower time-to-first-token, total latency, or settled cost in every cell.

Capability evidence

Transport success and functional success are separate. A 2xx response that does not satisfy the requested schema or tool contract is not a functional pass.

Platform contracts: hosted open models use automatic best-effort exact-prefix prompt caching. DigitalOcean batch inference does not support these open-source hosted models. Those documented product states are not inferred from malformed-parameter probes.

Official product sources (verified 29 August 2026): prompt caching; batch limits; model catalog and feature matrix. Product documentation can change after the measurement date.

Endpoint	Response Format	Tools	Parallel Tool Calls	Vision	Automatic Prompt Cache	Batch Open Models
DeepSeek V4 Flash	not measured	passed 2/2	passed 2/2	not listed for vision; probe inconclusive	automatic; repeat-prefix reads seen	unsupported by product contract
Gemma 4 31B	not measured	passed 2/2	passed 2/2	not listed for vision; probe 1/1	automatic; no reads seen	unsupported by product contract
GLM 5.2	failed 0/2	passed 2/2	passed 2/2	docs: text only; probe inconclusive	automatic; repeat-prefix reads seen	unsupported by product contract
Kimi K3	not measured	failed 0/2	degraded 1/2	docs: image; probe 0/18	automatic; repeat-prefix reads seen	unsupported by product contract
MiniMax M2.5	not measured	degraded 1/2	degraded 1/2	not listed for vision; probe inconclusive	automatic; repeat-prefix reads seen	unsupported by product contract
MiMo V2.5 Pro	passed 2/2	passed 2/2	passed 2/2	docs: text only; probe inconclusive	automatic; reads not repeat-specific	unsupported by product contract
Nemotron 3 Ultra 550B	not measured	passed 2/2	degraded 1/2	not listed for vision; probe inconclusive	automatic; no reads seen	unsupported by product contract
Nemotron 3 Super 120B	not measured	not scored	not scored	not listed for vision; probe inconclusive	automatic; repeat-prefix reads seen	unsupported by product contract
GPT-OSS 120B	not measured	passed 2/2	degraded 1/2	not listed for vision; probe inconclusive	automatic; reads not repeat-specific	unsupported by product contract
Qwen3.5 397B A17B	not measured	not scored	not scored	not listed for vision; probe 0/1	automatic; repeat-prefix reads seen	unsupported by product contract
Qwen3.8 Max	passed 2/2	passed 2/2	passed 2/2	docs: image/video; probe 18/18	automatic; repeat-prefix reads seen	unsupported by product contract

Observed context boundaries

Accepted probes are not silently upgraded to exact maxima. Right-censored means the endpoint accepted through the largest tested value; interval-censored means the boundary lies between a measured accept and reject. Timeouts, 429s, and 5xx responses are inconclusive for capability.

Endpoint	Dimension	Documented	Measured finding	Boundary state
DeepSeek V4 Flash	prompt context window	1,048,576	Retrieval-valid through 2,722 estimated tokens	right censored
Gemma 4 31B	prompt context window	256,000	Retrieval-valid through 251,873 estimated tokens	right censored
GLM 5.2	prompt context window	262,144	Retrieval-valid through 182,823 estimated tokens	right censored
Kimi K3	prompt context window	not documented	Retrieval-valid through 158,411 estimated tokens	right censored
MiniMax M2.5	prompt context window	65,536	Retrieval-valid through 166,688 estimated tokens	right censored
MiMo V2.5 Pro	prompt context window	262,144	Retrieval-valid through 245,518 estimated tokens	right censored
Nemotron 3 Ultra 550B	prompt context window	131,072	Retrieval-valid through 76,667 estimated tokens	right censored
Nemotron 3 Super 120B	prompt context window	1,000,000	Retrieval-valid through 76,667 estimated tokens	interval censored
GPT-OSS 120B	prompt context window	128,000	Retrieval-valid through 66,717 estimated tokens	right censored
Qwen3.5 397B A17B	prompt context window	131,072	Mixed accept/reject order; no numeric boundary	nonmonotonic inconclusive
Qwen3.8 Max	prompt context window	1,000,000	Retrieval-valid through 1,004,307 estimated tokens	right censored

Observed long-output responses

These probes answer a practical question: how long a response did the endpoint actually complete in this campaign? They do not silently turn that response length into a maximum. A model can stop because its answer is complete or it emits an end token; establishing an exact limit requires a measured accept/reject boundary.

Endpoint	Frozen catalog value	Largest response observed	What this establishes
DeepSeek V4 Flash	1,048,576	10,020 generated tokens	At least this response length; not an exact maximum
Gemma 4 31B	8,192	947 generated tokens	At least this response length; not an exact maximum
GLM 5.2	262,144	4,096 generated tokens	At least this response length; not an exact maximum
Kimi K3	not documented	4,096 generated tokens	At least this response length; not an exact maximum
MiniMax M2.5	65,536	4,096 generated tokens	At least this response length; not an exact maximum
MiMo V2.5 Pro	262,144	4,096 generated tokens	At least this response length; not an exact maximum
Nemotron 3 Ultra 550B	131,072	4,096 generated tokens	At least this response length; not an exact maximum
Nemotron 3 Super 120B	32,768	4,096 generated tokens	At least this response length; not an exact maximum
GPT-OSS 120B	4,096	4,096 generated tokens	At least this response length; not an exact maximum
Qwen3.5 397B A17B	131,072	5,632 generated tokens	At least this response length; not an exact maximum
Qwen3.8 Max	262,144	10,380 generated tokens	At least this response length; not an exact maximum

Tested	Registered long-output probes from the static/capability campaign; one largest completed response retained per endpoint.
Shows	A measured lower bound on completed response length for the tested request, next to the frozen catalog value used by the runner.
Does not show	The exact maximum output, a guarantee that every request reaches this length, or current catalog metadata after 29 August 2026.

Operational implications

Use retry with randomized exponential backoff, per-endpoint concurrency controls, and circuit breaking. Treat 429 as a capacity signal, not an instruction to replay immediately. Do not retry malformed 400-class requests blindly. Preserve idempotency and observe both request count and tokens per minute because the binding quota can vary with workload.

Endpoint-specific cautions from this run

Qwen3.5 397B A17B: 72/112 successes (64.3%); 30 rate limits and 10 timeouts. Its short/long cell completed only 11/28 requests. Operate with conservative concurrency and alert on both 429 and long-tail latency.

Nemotron 3 Super 120B: 62/112 successes (55.4%); 47 rate limits and 3 timeouts. Failures were concentrated late in the six-hour run. Production use needs strict adaptive throttling and a fallback path.

Gemma 4 31B: 105/112 successes (93.8%). Reliability was substantially better than the two endpoints above, but not perfect; retain backoff and monitor protocol-level failures in addition to HTTP status.

For the eight 112/112 endpoints, the correct conclusion is narrower: they were reliable at this registered low offered load during this run. Use the workload-specific adaptive-load and fixed-rate pages before setting production concurrency.

Reproducibility and provenance

The benchmark runner is open-loop, parallel, idempotent, and receipt-backed. Requests remain attached to an exact endpoint, recipe, phase, panel, and source commit. The public package contains aggregate tables and audit manifests only; credentials, prompts, generated outputs, response bodies, and raw headers are excluded.

Evidence layer	Exact source	What it contributes
Static / capability	do-complete-20260827-r1; terminal SHA256SUMS 0b2a4772...	Caching, capabilities, context/output, quality, baseline latency
Adaptive load	do-capacity-20260828-r2 plus completed gap cells from do-six-hour-variation-20260828-r1	Endpoint-by-workload confirmed bounds, brackets, and measured lowest-rate failures
Fixed-rate stability	do-direct-soak-20260823-r1	Four-block two-minute stability and recovery evidence
Six-hour variation	do-six-hour-variation-20260828-r1; SHA256SUMS 9d3ea8a161ce...	7/7 hourly panels; 1,232/1,232 core requests; stable versus fresh-prefix matched design

Six-hour immutable identities: source commit 603061da297c5422a7ba1750a110d80bf26b4757; campaign hash 50218753aa9404e0946bbbc04692f95986df6042ad57a54cf19ea467a11c19de; terminal ledger SHA-256 b78194ef68ba9dbdbec9f6000917a266231b6b915722da4f4429722e894669dc.

Machine-readable companions include the capacity table, fixed-rate cell/block tables, capability and boundary tables, six-hour panel/across-panel/cache-effect tables, outlier audit, coverage matrix, and recursive publication-safety result.

DeepSeek V4 Flash

LOW-LOAD RELIABILITY PASS	A strong candidate for a monitored pilot at the measured low load. Size production traffic from the exact adaptive-load recipe below.

Provider	Documented context	Documented max output	Published token price
DeepSeek	1,048,576 tokens	1,048,576 tokens	$0.080 / $0.252 per 1M input / output

Six-hour low-load result: **112/112** successful (100.0%; Wilson 95% CI 96.7-100.0%). Every workload below has 14 or 28 requests across seven hourly panels; 429s and timeouts count as measured failures.

Traffic-handling evidence

Exact recipe	Adaptive load search	Two-minute fixed-rate test
Short prompt / short answer	**at least 32 req/s (tested lower bound)**	**passed**
100K prompt / short answer	**at least 0.5 req/s (tested lower bound)**	**fixed-rate test not run for this exact recipe**
32K prompt / short answer	**adaptive search not run for this exact recipe**	**measured failure**
Short prompt / long answer	**at least 1 req/s (tested lower bound)**	**measured failure**
Mixed application traffic	**0.067-0.2 req/s bracket**	**measured failure**

Six-hour behavior by matched low-load workload

Low-load workload	Request success (Wilson 95%)	Panel-median latency: stable / fresh, s [95%]	Eligible decode rate: stable / fresh, tok/s [95%]
Short / short	28/28 (100.0%; 95% 87.9-100.0%)	0.7 [0.6-0.7] / 0.6 [0.6-1.0] s	not estimable / not estimable tok/s
100K / short	28/28 (100.0%; 95% 87.9-100.0%)	7.1 [6.4-8.3] / 7.4 [6.8-7.6] s	40.6 [17.2-54.9] / 51.1 [48.9-53.3] tok/s
Short / long	28/28 (100.0%; 95% 87.9-100.0%)	77.4 [74.5-83.8] / 83.3 [74.1-85.3] s	50.6 [48.7-54.3] / 49.7 [48.5-55.6] tok/s
Mixed: structured	14/14 (100.0%; 95% 78.5-100.0%)	3.3 [2.9-4.5] / 3.1 [2.9-3.8] s	54.7 [48.6-55.9] / 55.0 [53.3-55.7] tok/s
Mixed: 100K input	14/14 (100.0%; 95% 78.5-100.0%)	6.5 [1.7-8.9] / 7.0 [6.8-8.8] s	52.3 (no interval) / 28.5 [3.4-53.6] tok/s

Stable / fresh means repeated exact token prefix / panel-unique fresh prefix. Latency and decode intervals resample the seven hourly panels; not estimable means the strict timestamp, complete-usage, one-second decode-window, and 16-output-token rules were not met.

Capabilities, caching, and context boundary

Capability	Measured / documented state	Capability	Measured / documented state
response format	not measured	tools	passed (2/2)
parallel tool calls	passed (2/2)	vision	not documented as vision-capable; live probe inconclusive
automatic prompt cache	automatic (docs); exact-prefix reuse observed	batch open models	unsupported (documented)

Measured context: retrieval-valid prompt accepted through 2,722 estimated tokens (right censored); combined prompt-plus-output target transport-accepted through 1,044,925 estimated tokens; retrieval was not verified. Estimated tokens can differ from the provider tokenizer.

Product-contract sources: model catalog, pricing, prompt caching, and batch limits; verified 29 August 2026.

Gemma 4 31B

PILOT WITH FALLBACK	Usable for a monitored pilot at the measured low load. Keep retry/backoff and a fallback; do not infer high-load capacity from this result. Failure mix: 5 rate-limited, 1 timed out, 1 server errors.

Provider	Documented context	Documented max output	Published token price
Google	256,000 tokens	8,192 tokens	$0.18 / $0.50 per 1M input / output

Six-hour low-load result: **105/112** successful (93.8%; Wilson 95% CI 87.7-96.9%). Every workload below has 14 or 28 requests across seven hourly panels; 429s and timeouts count as measured failures.

Traffic-handling evidence

Exact recipe	Adaptive load search	Two-minute fixed-rate test
Short prompt / short answer	**0.25-2.5 req/s bracket**	**passed**
100K prompt / short answer	**below 0.5 req/s (failed lowest test)**	**fixed-rate test not run for this exact recipe**
32K prompt / short answer	**adaptive search not run for this exact recipe**	**measured failure**
Short prompt / long answer	**at least 1 req/s (tested lower bound)**	**measured failure**
Mixed application traffic	**at least 2 req/s (tested lower bound)**	**measured failure**

Six-hour behavior by matched low-load workload

Low-load workload	Request success (Wilson 95%)	Panel-median latency: stable / fresh, s [95%]	Eligible decode rate: stable / fresh, tok/s [95%]
Short / short	27/28 (96.4%; 95% 82.3-99.4%)	1.1 [0.6-11.1] / 1.2 [0.7-27.6] s	not estimable / not estimable tok/s
100K / short	26/28 (92.9%; 95% 77.4-98.0%)	138.8 [98.4-154.4] / 55.4 [52.9-60.9] s	25.7 [15.4-28.8] / 21.3 [11.0-27.9] tok/s
Short / long	25/28 (89.3%; 95% 72.8-96.3%)	78.8 [50.0-145.5] / 66.4 [40.7-97.7] s	32.7 [22.9-44.7] / 37.0 [25.1-50.2] tok/s
Mixed: structured	13/14 (92.9%; 95% 68.5-98.7%)	2.6 [2.0-2.7] / 3.2 [2.6-35.5] s	45.0 [42.6-53.1] / 41.0 [10.9-43.6] tok/s
Mixed: 100K input	14/14 (100.0%; 95% 78.5-100.0%)	53.4 [52.4-55.1] / 58.4 [53.9-71.9] s	27.3 [24.8-31.7] / 7.3 [3.7-19.5] tok/s

Stable / fresh means repeated exact token prefix / panel-unique fresh prefix. Latency and decode intervals resample the seven hourly panels; not estimable means the strict timestamp, complete-usage, one-second decode-window, and 16-output-token rules were not met.

Capabilities, caching, and context boundary

Capability	Measured / documented state	Capability	Measured / documented state
response format	not measured	tools	passed (2/2)
parallel tool calls	passed (2/2)	vision	not documented as vision-capable; live probe passed (1/1)
automatic prompt cache	automatic (docs); no cache reads observed in this panel	batch open models	unsupported (documented)

Measured context: retrieval-valid prompt accepted through 251,873 estimated tokens (right censored); combined prompt-plus-output target transport-accepted through 130,296 estimated tokens; retrieval was not verified. Estimated tokens can differ from the provider tokenizer.

Product-contract sources: model catalog, pricing, prompt caching, and batch limits; verified 29 August 2026.

GLM 5.2

LOW-LOAD RELIABILITY PASS	A strong candidate for a monitored pilot at the measured low load. Size production traffic from the exact adaptive-load recipe below.

Provider	Documented context	Documented max output	Published token price
Z.ai	262,144 tokens	262,144 tokens	$0.70 / $2.20 per 1M input / output

Six-hour low-load result: **112/112** successful (100.0%; Wilson 95% CI 96.7-100.0%). Every workload below has 14 or 28 requests across seven hourly panels; 429s and timeouts count as measured failures.

Traffic-handling evidence

Exact recipe	Adaptive load search	Two-minute fixed-rate test
Short prompt / short answer	**below 0.5 req/s (failed lowest test)**	**measured failure**
100K prompt / short answer	**below 0.5 req/s (failed lowest test)**	**fixed-rate test not run for this exact recipe**
32K prompt / short answer	**adaptive search not run for this exact recipe**	**measured failure**
Short prompt / long answer	**below 0.5 req/s (failed lowest test)**	**measured failure**
Mixed application traffic	**0.067-0.2 req/s bracket**	**measured failure**

Six-hour behavior by matched low-load workload

Low-load workload	Request success (Wilson 95%)	Panel-median latency: stable / fresh, s [95%]	Eligible decode rate: stable / fresh, tok/s [95%]
Short / short	28/28 (100.0%; 95% 87.9-100.0%)	1.3 [1.2-1.4] / 1.2 [1.2-1.4] s	not estimable / not estimable tok/s
100K / short	28/28 (100.0%; 95% 87.9-100.0%)	6.5 [2.5-7.0] / 7.0 [6.7-7.2] s	not estimable / not estimable tok/s
Short / long	28/28 (100.0%; 95% 87.9-100.0%)	26.6 [23.5-28.6] / 27.0 [26.3-28.0] s	not estimable / not estimable tok/s
Mixed: structured	14/14 (100.0%; 95% 78.5-100.0%)	4.0 [3.7-4.1] / 3.1 [1.9-3.6] s	not estimable / not estimable tok/s
Mixed: 100K input	14/14 (100.0%; 95% 78.5-100.0%)	6.6 [5.7-7.8] / 6.8 [6.6-7.2] s	not estimable / not estimable tok/s

Stable / fresh means repeated exact token prefix / panel-unique fresh prefix. Latency and decode intervals resample the seven hourly panels; not estimable means the strict timestamp, complete-usage, one-second decode-window, and 16-output-token rules were not met.

Capabilities, caching, and context boundary

Capability	Measured / documented state	Capability	Measured / documented state
response format	failed (0/2)	tools	passed (2/2)
parallel tool calls	passed (2/2)	vision	text-only documented; live probe inconclusive
automatic prompt cache	automatic (docs); exact-prefix reuse observed	batch open models	unsupported (documented)

Measured context: retrieval-valid prompt accepted through 182,823 estimated tokens (right censored); combined prompt-plus-output target transport-accepted through 265,289 estimated tokens; retrieval was not verified. Estimated tokens can differ from the provider tokenizer.

Product-contract sources: model catalog, pricing, prompt caching, and batch limits; verified 29 August 2026.

Kimi K3

LOW-LOAD RELIABILITY PASS	A strong candidate for a monitored pilot at the measured low load. Size production traffic from the exact adaptive-load recipe below.

Provider	Documented context	Documented max output	Published token price
Moonshot AI	not documented	not documented	$2.85 / $14.25 per 1M input / output

Six-hour low-load result: **112/112** successful (100.0%; Wilson 95% CI 96.7-100.0%). Every workload below has 14 or 28 requests across seven hourly panels; 429s and timeouts count as measured failures.

Traffic-handling evidence

Exact recipe	Adaptive load search	Two-minute fixed-rate test
Short prompt / short answer	**below 0.5 req/s (failed lowest test)**	**measured failure**
100K prompt / short answer	**adaptive search not run for this exact recipe**	**fixed-rate test not run for this exact recipe**
32K prompt / short answer	**at least 0.8 req/s (tested lower bound)**	**measured failure**
Short prompt / long answer	**below 0.5 req/s (failed lowest test)**	**measured failure**
Mixed application traffic	**below 0.5 req/s (failed lowest test)**	**measured failure**

Six-hour behavior by matched low-load workload

Low-load workload	Request success (Wilson 95%)	Panel-median latency: stable / fresh, s [95%]	Eligible decode rate: stable / fresh, tok/s [95%]
Short / short	28/28 (100.0%; 95% 87.9-100.0%)	3.5 [3.4-4.1] / 3.7 [2.8-4.4] s	not estimable / not estimable tok/s
100K / short	28/28 (100.0%; 95% 87.9-100.0%)	10.3 [9.7-10.7] / 9.8 [9.7-10.6] s	not estimable / not estimable tok/s
Short / long	28/28 (100.0%; 95% 87.9-100.0%)	111.7 [106.9-114.0] / 90.5 [62.0-107.2] s	not estimable / not estimable tok/s
Mixed: structured	14/14 (100.0%; 95% 78.5-100.0%)	10.5 [9.3-16.7] / 12.7 [11.8-14.4] s	not estimable / not estimable tok/s
Mixed: 100K input	14/14 (100.0%; 95% 78.5-100.0%)	9.8 [9.4-10.4] / 10.4 [9.8-10.6] s	not estimable / not estimable tok/s

Stable / fresh means repeated exact token prefix / panel-unique fresh prefix. Latency and decode intervals resample the seven hourly panels; not estimable means the strict timestamp, complete-usage, one-second decode-window, and 16-output-token rules were not met.

Capabilities, caching, and context boundary

Capability	Measured / documented state	Capability	Measured / documented state
response format	not measured	tools	failed (0/2)
parallel tool calls	degraded (1/2)	vision	image documented; live probe failed (0/18)
automatic prompt cache	automatic (docs); exact-prefix reuse observed	batch open models	unsupported (documented)

Measured context: retrieval-valid prompt accepted through 158,411 estimated tokens (right censored); combined prompt-plus-output target transport-accepted through 53,633 estimated tokens; retrieval was not verified. Estimated tokens can differ from the provider tokenizer.

Product-contract sources: model catalog, pricing, prompt caching, and batch limits; verified 29 August 2026.

MiniMax M2.5

LOW-LOAD RELIABILITY PASS	A strong candidate for a monitored pilot at the measured low load. Size production traffic from the exact adaptive-load recipe below.

Provider	Documented context	Documented max output	Published token price
MiniMax	65,536 tokens	65,536 tokens	$0.30 / $1.20 per 1M input / output

Six-hour low-load result: **112/112** successful (100.0%; Wilson 95% CI 96.7-100.0%). Every workload below has 14 or 28 requests across seven hourly panels; 429s and timeouts count as measured failures.

Traffic-handling evidence

Exact recipe	Adaptive load search	Two-minute fixed-rate test
Short prompt / short answer	**below 0.5 req/s (failed lowest test)**	**measured failure**
100K prompt / short answer	**adaptive search not run for this exact recipe**	**fixed-rate test not run for this exact recipe**
32K prompt / short answer	**0.47-0.8 req/s bracket**	**measured failure**
Short prompt / long answer	**at least 0.47 req/s (tested lower bound)**	**measured failure**
Mixed application traffic	**at least 1 req/s (tested lower bound)**	**measured failure**

Six-hour behavior by matched low-load workload

Low-load workload	Request success (Wilson 95%)	Panel-median latency: stable / fresh, s [95%]	Eligible decode rate: stable / fresh, tok/s [95%]
Short / short	28/28 (100.0%; 95% 87.9-100.0%)	2.2 [2.1-2.2] / 2.4 [2.0-3.2] s	not estimable / not estimable tok/s
100K / short	28/28 (100.0%; 95% 87.9-100.0%)	3.3 [2.9-4.8] / 6.1 [6.0-7.1] s	not estimable / not estimable tok/s
Short / long	28/28 (100.0%; 95% 87.9-100.0%)	57.2 [52.6-69.5] / 57.3 [55.4-67.2] s	not estimable / not estimable tok/s
Mixed: structured	14/14 (100.0%; 95% 78.5-100.0%)	7.5 [6.8-8.6] / 6.9 [6.7-7.6] s	not estimable / not estimable tok/s
Mixed: 50K input	14/14 (100.0%; 95% 78.5-100.0%)	3.0 [2.9-5.4] / 6.1 [5.7-6.4] s	not estimable / not estimable tok/s

Stable / fresh means repeated exact token prefix / panel-unique fresh prefix. Latency and decode intervals resample the seven hourly panels; not estimable means the strict timestamp, complete-usage, one-second decode-window, and 16-output-token rules were not met.

Capabilities, caching, and context boundary

Capability	Measured / documented state	Capability	Measured / documented state
response format	not measured	tools	degraded (1/2)
parallel tool calls	degraded (1/2)	vision	not documented as vision-capable; live probe inconclusive
automatic prompt cache	automatic (docs); exact-prefix reuse observed	batch open models	unsupported (documented)

Measured context: retrieval-valid prompt accepted through 166,688 estimated tokens (right censored); combined prompt-plus-output target transport-accepted through 60,051 estimated tokens; retrieval was not verified. Estimated tokens can differ from the provider tokenizer.

Product-contract sources: model catalog, pricing, prompt caching, and batch limits; verified 29 August 2026.

MiMo V2.5 Pro

LOW-LOAD RELIABILITY PASS	A strong candidate for a monitored pilot at the measured low load. Size production traffic from the exact adaptive-load recipe below.

Provider	Documented context	Documented max output	Published token price
Xiaomi	262,144 tokens	262,144 tokens	$0.40 / $1.50 per 1M input / output

Six-hour low-load result: **112/112** successful (100.0%; Wilson 95% CI 96.7-100.0%). Every workload below has 14 or 28 requests across seven hourly panels; 429s and timeouts count as measured failures.

Traffic-handling evidence

Exact recipe	Adaptive load search	Two-minute fixed-rate test
Short prompt / short answer	**at least 8 req/s (tested lower bound)**	**measured failure**
100K prompt / short answer	**below 0.5 req/s (failed lowest test)**	**fixed-rate test not run for this exact recipe**
32K prompt / short answer	**adaptive search not run for this exact recipe**	**measured failure**
Short prompt / long answer	**at least 1 req/s (tested lower bound)**	**measured failure**
Mixed application traffic	**below 0.5 req/s (failed lowest test)**	**measured failure**

Six-hour behavior by matched low-load workload

Low-load workload	Request success (Wilson 95%)	Panel-median latency: stable / fresh, s [95%]	Eligible decode rate: stable / fresh, tok/s [95%]
Short / short	28/28 (100.0%; 95% 87.9-100.0%)	7.5 [5.0-13.6] / 7.1 [5.0-11.7] s	not estimable / not estimable tok/s
100K / short	28/28 (100.0%; 95% 87.9-100.0%)	12.5 [10.3-15.4] / 11.2 [8.9-16.2] s	not estimable / not estimable tok/s
Short / long	28/28 (100.0%; 95% 87.9-100.0%)	92.4 [80.3-134.2] / 115.0 [59.3-162.4] s	not estimable / not estimable tok/s
Mixed: structured	14/14 (100.0%; 95% 78.5-100.0%)	11.9 [8.4-20.7] / 19.7 [16.1-28.2] s	not estimable / not estimable tok/s
Mixed: 100K input	14/14 (100.0%; 95% 78.5-100.0%)	13.3 [7.4-17.9] / 14.8 [10.5-16.7] s	not estimable / not estimable tok/s

Stable / fresh means repeated exact token prefix / panel-unique fresh prefix. Latency and decode intervals resample the seven hourly panels; not estimable means the strict timestamp, complete-usage, one-second decode-window, and 16-output-token rules were not met.

Capabilities, caching, and context boundary

Capability	Measured / documented state	Capability	Measured / documented state
response format	passed (2/2)	tools	passed (2/2)
parallel tool calls	passed (2/2)	vision	text-only documented; live probe inconclusive
automatic prompt cache	automatic (docs); cache reads not specific to exact repeats	batch open models	unsupported (documented)

Measured context: retrieval-valid prompt accepted through 245,518 estimated tokens (right censored); combined prompt-plus-output target transport-accepted through 246,272 estimated tokens; retrieval was not verified. Estimated tokens can differ from the provider tokenizer.

Product-contract sources: model catalog, pricing, prompt caching, and batch limits; verified 29 August 2026.

Nemotron 3 Ultra 550B

LOW-LOAD RELIABILITY PASS	A strong candidate for a monitored pilot at the measured low load. Size production traffic from the exact adaptive-load recipe below.

Provider	Documented context	Documented max output	Published token price
NVIDIA	131,072 tokens	131,072 tokens	$0.90 / $1.70 per 1M input / output

Six-hour low-load result: **112/112** successful (100.0%; Wilson 95% CI 96.7-100.0%). Every workload below has 14 or 28 requests across seven hourly panels; 429s and timeouts count as measured failures.

Traffic-handling evidence

Exact recipe	Adaptive load search	Two-minute fixed-rate test
Short prompt / short answer	**at least 2 req/s (tested lower bound)**	**measured failure**
100K prompt / short answer	**below 0.5 req/s (failed lowest test)**	**fixed-rate test not run for this exact recipe**
32K prompt / short answer	**adaptive search not run for this exact recipe**	**measured failure**
Short prompt / long answer	**at least 1 req/s (tested lower bound)**	**measured failure**
Mixed application traffic	**at least 0.2 req/s (tested lower bound)**	**measured failure**

Six-hour behavior by matched low-load workload

Low-load workload	Request success (Wilson 95%)	Panel-median latency: stable / fresh, s [95%]	Eligible decode rate: stable / fresh, tok/s [95%]
Short / short	28/28 (100.0%; 95% 87.9-100.0%)	3.8 [1.7-5.9] / 1.7 [1.4-2.7] s	not estimable / not estimable tok/s
100K / short	28/28 (100.0%; 95% 87.9-100.0%)	9.6 [9.1-14.5] / 9.1 [8.7-11.3] s	not estimable / not estimable tok/s
Short / long	28/28 (100.0%; 95% 87.9-100.0%)	50.9 [42.4-53.9] / 53.9 [51.7-60.9] s	not estimable / not estimable tok/s
Mixed: structured	14/14 (100.0%; 95% 78.5-100.0%)	3.2 [2.7-4.7] / 3.2 [3.0-3.6] s	not estimable / not estimable tok/s
Mixed: 100K input	14/14 (100.0%; 95% 78.5-100.0%)	9.1 [8.9-9.6] / 9.1 [8.9-9.8] s	not estimable / not estimable tok/s

Stable / fresh means repeated exact token prefix / panel-unique fresh prefix. Latency and decode intervals resample the seven hourly panels; not estimable means the strict timestamp, complete-usage, one-second decode-window, and 16-output-token rules were not met.

Capabilities, caching, and context boundary

Capability	Measured / documented state	Capability	Measured / documented state
response format	not measured	tools	passed (2/2)
parallel tool calls	degraded (1/2)	vision	not documented as vision-capable; live probe inconclusive
automatic prompt cache	automatic (docs); no cache reads observed in this panel	batch open models	unsupported (documented)

Measured context: retrieval-valid prompt accepted through 76,667 estimated tokens (right censored); combined prompt-plus-output target transport-accepted through 126,176 estimated tokens; retrieval was not verified. Estimated tokens can differ from the provider tokenizer.

Product-contract sources: model catalog, pricing, prompt caching, and batch limits; verified 29 August 2026.

Nemotron 3 Super 120B

PRODUCTION CAUTION	Do not use as the only route yet. Start with conservative concurrency, adaptive backoff, a circuit breaker, and a tested fallback. Failure mix: 47 rate-limited, 3 timed out, 0 server errors.

Provider	Documented context	Documented max output	Published token price
NVIDIA	1,000,000 tokens	32,768 tokens	$0.30 / $0.65 per 1M input / output

Six-hour low-load result: **62/112** successful (55.4%; Wilson 95% CI 46.1-64.2%). Every workload below has 14 or 28 requests across seven hourly panels; 429s and timeouts count as measured failures.

Traffic-handling evidence

Exact recipe	Adaptive load search	Two-minute fixed-rate test
Short prompt / short answer	**below 0.5 req/s (failed lowest test)**	**measured failure**
100K prompt / short answer	**below 0.5 req/s (failed lowest test)**	**fixed-rate test not run for this exact recipe**
32K prompt / short answer	**adaptive search not run for this exact recipe**	**measured failure**
Short prompt / long answer	**below 0.5 req/s (failed lowest test)**	**measured failure**
Mixed application traffic	**below 0.5 req/s (failed lowest test)**	**measured failure**

Six-hour behavior by matched low-load workload

Low-load workload	Request success (Wilson 95%)	Panel-median latency: stable / fresh, s [95%]	Eligible decode rate: stable / fresh, tok/s [95%]
Short / short	19/28 (67.9%; 95% 49.3-82.1%)	5.4 [3.9-10.9] / 7.5 [3.9-20.3] s	not estimable / not estimable tok/s
100K / short	15/28 (53.6%; 95% 35.8-70.5%)	14.5 [11.8-20.1] / 17.5 [15.2-32.6] s	not estimable / not estimable tok/s
Short / long	14/28 (50.0%; 95% 32.6-67.4%)	313.9 [196.0-353.0] / 211.1 [194.1-326.4] s	not estimable / not estimable tok/s
Mixed: structured	8/14 (57.1%; 95% 32.6-78.6%)	13.9 [12.9-14.8] / 35.9 [16.0-51.1] s	not estimable / not estimable tok/s
Mixed: 100K input	6/14 (42.9%; 95% 21.4-67.4%)	12.4 [10.0-17.0] / 17.0 [16.6-17.9] s	not estimable / not estimable tok/s

Stable / fresh means repeated exact token prefix / panel-unique fresh prefix. Latency and decode intervals resample the seven hourly panels; not estimable means the strict timestamp, complete-usage, one-second decode-window, and 16-output-token rules were not met.

Capabilities, caching, and context boundary

Capability	Measured / documented state	Capability	Measured / documented state
response format	not measured	tools	not scored
parallel tool calls	not scored	vision	not documented as vision-capable; live probe inconclusive
automatic prompt cache	automatic (docs); exact-prefix reuse observed	batch open models	unsupported (documented)

Measured context: retrieval-valid prompt accepted through 76,667 estimated tokens (interval censored). Estimated tokens can differ from the provider tokenizer.

Product-contract sources: model catalog, pricing, prompt caching, and batch limits; verified 29 August 2026.

GPT-OSS 120B

LOW-LOAD RELIABILITY PASS	A strong candidate for a monitored pilot at the measured low load. Size production traffic from the exact adaptive-load recipe below.

Provider	Documented context	Documented max output	Published token price
OpenAI	128,000 tokens	4,096 tokens	$0.10 / $0.70 per 1M input / output

Six-hour low-load result: **112/112** successful (100.0%; Wilson 95% CI 96.7-100.0%). Every workload below has 14 or 28 requests across seven hourly panels; 429s and timeouts count as measured failures.

Traffic-handling evidence

Exact recipe	Adaptive load search	Two-minute fixed-rate test
Short prompt / short answer	**at least 8 req/s (tested lower bound)**	**measured failure**
100K prompt / short answer	**below 0.5 req/s (failed lowest test)**	**fixed-rate test not run for this exact recipe**
32K prompt / short answer	**adaptive search not run for this exact recipe**	**measured failure**
Short prompt / long answer	**at least 1 req/s (tested lower bound)**	**measured failure**
Mixed application traffic	**0.067-0.2 req/s bracket**	**measured failure**

Six-hour behavior by matched low-load workload

Low-load workload	Request success (Wilson 95%)	Panel-median latency: stable / fresh, s [95%]	Eligible decode rate: stable / fresh, tok/s [95%]
Short / short	28/28 (100.0%; 95% 87.9-100.0%)	2.9 [2.7-3.3] / 2.4 [2.3-3.0] s	not estimable / not estimable tok/s
100K / short	28/28 (100.0%; 95% 87.9-100.0%)	12.6 [12.0-13.1] / 12.6 [12.0-13.0] s	not estimable / not estimable tok/s
Short / long	28/28 (100.0%; 95% 87.9-100.0%)	124.1 [120.7-125.9] / 118.3 [111.0-120.9] s	not estimable / not estimable tok/s
Mixed: structured	14/14 (100.0%; 95% 78.5-100.0%)	10.7 [8.8-12.0] / 10.2 [8.9-12.4] s	not estimable / not estimable tok/s
Mixed: 100K input	14/14 (100.0%; 95% 78.5-100.0%)	12.7 [11.9-13.6] / 12.3 [10.6-13.0] s	not estimable / not estimable tok/s

Stable / fresh means repeated exact token prefix / panel-unique fresh prefix. Latency and decode intervals resample the seven hourly panels; not estimable means the strict timestamp, complete-usage, one-second decode-window, and 16-output-token rules were not met.

Capabilities, caching, and context boundary

Capability	Measured / documented state	Capability	Measured / documented state
response format	not measured	tools	passed (2/2)
parallel tool calls	degraded (1/2)	vision	not documented as vision-capable; live probe inconclusive
automatic prompt cache	automatic (docs); cache reads not specific to exact repeats	batch open models	unsupported (documented)

Measured context: retrieval-valid prompt accepted through 66,717 estimated tokens (right censored); combined prompt-plus-output target transport-accepted through 117,823 estimated tokens; retrieval was not verified. Estimated tokens can differ from the provider tokenizer.

Product-contract sources: model catalog, pricing, prompt caching, and batch limits; verified 29 August 2026.

Qwen3.5 397B A17B

PRODUCTION CAUTION	Do not use as the only route yet. Start with conservative concurrency, adaptive backoff, a circuit breaker, and a tested fallback. Failure mix: 30 rate-limited, 10 timed out, 0 server errors.

Provider	Documented context	Documented max output	Published token price
Alibaba	131,072 tokens	131,072 tokens	$0.55 / $3.50 per 1M input / output

Six-hour low-load result: **72/112** successful (64.3%; Wilson 95% CI 55.1-72.6%). Every workload below has 14 or 28 requests across seven hourly panels; 429s and timeouts count as measured failures.

Traffic-handling evidence

Exact recipe	Adaptive load search	Two-minute fixed-rate test
Short prompt / short answer	**below 0.5 req/s (failed lowest test)**	**measured failure**
100K prompt / short answer	**below 0.5 req/s (failed lowest test)**	**fixed-rate test not run for this exact recipe**
32K prompt / short answer	**adaptive search not run for this exact recipe**	**measured failure**
Short prompt / long answer	**below 0.5 req/s (failed lowest test)**	**measured failure**
Mixed application traffic	**below 0.5 req/s (failed lowest test)**	**measured failure**

Six-hour behavior by matched low-load workload

Low-load workload	Request success (Wilson 95%)	Panel-median latency: stable / fresh, s [95%]	Eligible decode rate: stable / fresh, tok/s [95%]
Short / short	18/28 (64.3%; 95% 45.8-79.3%)	44.0 [18.0-79.8] / 23.7 [7.8-50.2] s	not estimable / not estimable tok/s
100K / short	20/28 (71.4%; 95% 52.9-84.7%)	47.4 [31.3-104.1] / 60.7 [53.7-113.5] s	not estimable / not estimable tok/s
Short / long	11/28 (39.3%; 95% 23.6-57.6%)	300.2 [229.4-338.4] / 191.9 [89.6-268.1] s	not estimable / not estimable tok/s
Mixed: structured	13/14 (92.9%; 95% 68.5-98.7%)	85.2 [41.2-146.0] / 82.5 [52.2-157.1] s	not estimable / not estimable tok/s
Mixed: 100K input	10/14 (71.4%; 95% 45.4-88.3%)	80.4 [51.4-108.8] / 64.2 [54.4-99.1] s	not estimable / not estimable tok/s

Stable / fresh means repeated exact token prefix / panel-unique fresh prefix. Latency and decode intervals resample the seven hourly panels; not estimable means the strict timestamp, complete-usage, one-second decode-window, and 16-output-token rules were not met.

Capabilities, caching, and context boundary

Capability	Measured / documented state	Capability	Measured / documented state
response format	not measured	tools	not scored
parallel tool calls	not scored	vision	not documented as vision-capable; live probe failed (0/1)
automatic prompt cache	automatic (docs); exact-prefix reuse observed	batch open models	unsupported (documented)

Measured context: retrieval-valid prompt accepted through 98,141 estimated tokens (nonmonotonic inconclusive); combined prompt-plus-output target transport-accepted through 70,999 estimated tokens; retrieval was not verified. Estimated tokens can differ from the provider tokenizer.

Product-contract sources: model catalog, pricing, prompt caching, and batch limits; verified 29 August 2026.

Qwen3.8 Max

LOW-LOAD RELIABILITY PASS	A strong candidate for a monitored pilot at the measured low load. Size production traffic from the exact adaptive-load recipe below.

Provider	Documented context	Documented max output	Published token price
Alibaba	1,000,000 tokens	262,144 tokens	$2.00 / $6.00 per 1M input / output

Six-hour low-load result: **112/112** successful (100.0%; Wilson 95% CI 96.7-100.0%). Every workload below has 14 or 28 requests across seven hourly panels; 429s and timeouts count as measured failures.

Traffic-handling evidence

Exact recipe	Adaptive load search	Two-minute fixed-rate test
Short prompt / short answer	**at least 8 req/s (tested lower bound)**	**passed**
100K prompt / short answer	**at least 0.5 req/s (tested lower bound)**	**fixed-rate test not run for this exact recipe**
32K prompt / short answer	**adaptive search not run for this exact recipe**	**measured failure**
Short prompt / long answer	**at least 1 req/s (tested lower bound)**	**measured failure**
Mixed application traffic	**at least 2 req/s (tested lower bound)**	**measured failure**

Six-hour behavior by matched low-load workload

Low-load workload	Request success (Wilson 95%)	Panel-median latency: stable / fresh, s [95%]	Eligible decode rate: stable / fresh, tok/s [95%]
Short / short	28/28 (100.0%; 95% 87.9-100.0%)	1.3 [1.1-1.7] / 1.8 [1.4-3.0] s	2.6 (no interval) / 2.4 (no interval) tok/s
100K / short	28/28 (100.0%; 95% 87.9-100.0%)	11.1 [6.8-11.8] / 10.7 [8.0-16.4] s	45.5 [23.3-60.7] / 56.2 [13.4-60.5] tok/s
Short / long	28/28 (100.0%; 95% 87.9-100.0%)	81.3 [67.2-101.9] / 102.5 [70.4-106.4] s	51.7 [41.6-65.4] / 40.6 [39.4-61.8] tok/s
Mixed: structured	14/14 (100.0%; 95% 78.5-100.0%)	8.7 [6.8-11.5] / 3.9 [3.7-11.9] s	35.9 [23.1-48.7] / 62.1 [17.0-85.2] tok/s
Mixed: 100K input	14/14 (100.0%; 95% 78.5-100.0%)	10.5 [3.5-12.5] / 10.7 [10.0-11.0] s	48.9 [20.9-65.0] / 54.3 [31.3-63.0] tok/s

Stable / fresh means repeated exact token prefix / panel-unique fresh prefix. Latency and decode intervals resample the seven hourly panels; not estimable means the strict timestamp, complete-usage, one-second decode-window, and 16-output-token rules were not met.

Capabilities, caching, and context boundary

Capability	Measured / documented state	Capability	Measured / documented state
response format	passed (2/2)	tools	passed (2/2)
parallel tool calls	passed (2/2)	vision	image/video documented; live probe passed (18/18)
automatic prompt cache	automatic (docs); exact-prefix reuse observed	batch open models	unsupported (documented)

Measured context: retrieval-valid prompt accepted through 1,004,307 estimated tokens (right censored); combined prompt-plus-output target transport-accepted through 1,002,293 estimated tokens; retrieval was not verified. Estimated tokens can differ from the provider tokenizer.

Product-contract sources: model catalog, pricing, prompt caching, and batch limits; verified 29 August 2026.